⌈Def. Let V be a vector space over a field F.
A function $f: V \times V \to F$ is called a bilinear form if: for all $\alpha_1, \alpha_2, \beta_1, \beta_2 \in V$, $c, d \in F$,

$$f(c\cdot\alpha_1+\alpha_2, d\cdot\beta_1+\beta_2) = cd\cdot f(\alpha_1,\beta_1) + c\cdot f(\alpha_1,\beta_2) + d\cdot f(\alpha_2,\beta_1) + f(\alpha_2,\beta_2).$$

⌈Def. Let V be a fin. dim vector space over F, and let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be a basis for V.
Given a bilinear form f, define the matrix of f in $\mathcal{B}$ as;

$$[f]_{\mathcal{B}} \overset{\text{def}}{=} (f(\alpha_i, \alpha_j))_{i,j}.$$

⌈Thm. Let V be a finite-dimensional vector space over F, and $\mathcal{B}$ be a basis.
Then, the space $L(V,V,F) = \{f: V\times V \to F \mid f \text{ is bilinear form}\}$ is isomorphic to $M_{n\times n}(F)$.
Proof. Define $\varphi: L(V,V,F) \to M_{n\times n}(F);\ f \mapsto [f]_{\mathcal{B}}$. Denote $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$.
Given $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$ and $\beta = y_1\alpha_1 + \cdots + y_n\alpha_n$,

$$f(\alpha,\beta) = \sum_{i=1}^n x_i\cdot f(\alpha_i,\beta) = \sum_{i=1}^n x_i \cdot \sum_{j=1}^n y_j \cdot f(\alpha_i,\alpha_j) = \sum_{i=1}^n\sum_{j=1}^n x_i y_j \cdot f(\alpha_i,\alpha_j).$$

Thus, for $X = [\alpha]_{\mathcal{B}}$ and $Y = [\beta]_{\mathcal{B}}$, and $A = [f]_{\mathcal{B}}$,

$$f(\alpha,\beta) = X^t A Y = [\alpha]_{\mathcal{B}}^t [f]_{\mathcal{B}} [\beta]_{\mathcal{B}}.$$

Therefore, the bilinear form f is uniquely determined by the representation $[f]_{\mathcal{B}}$.
This proves the φ is bijection. And, the φ is linear, because

$$(c\cdot f+g)(\alpha_i,\alpha_j) = c\cdot f(\alpha_i,\alpha_j) + g(\alpha_i,\alpha_j), \text{ for any } c\in F \text{ and } f,g \in L(V,V,F).$$

⌈Cor. If $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ is a basis for V, and $\mathcal{B}^* = \{f_1, \cdots, f_n\}$ is the dual basis,
then $f_{ij}(\alpha,\beta) \overset{\text{def}}{=} f_i(\alpha)\cdot f_j(\beta)$ $(1 \le i,j \le n)$ form a basis for V^{**}.